Sharing maps and the fixed-output locus

All results below are reproduced from the identified sources, with notation explained and scopes retained. This freeze does not add an independent mathematical recertification. PROVED denotes the supplied source-system status.

PROVED / TC-G1: coordinate model

Fix real $z > w$, $c = z-w$, and let $r = c_+(z,u)$, $b = c_+(w,v)$, $D = c_-(z,u) = c_-(w,v)$. Here $c_\pm(z,u) = (z-u^3 \pm u)/2$. Put $d = u-v$ and $\ell = u+v$. The sharing identity becomes

$$d(3\ell^2+d^2+4) = 4c,\ X = 12c/d - 4,\ Y = 36c\ell/d.$$

$$d = 12c/(X+4),\ \ell = Y/(3(X+4)),$$
$$u = (\ell+d)/2,\ v = (\ell-d)/2.$$

Substitution gives E_c: $Y^2 = X^3-48X-128-432c^2$, with discriminant $\Delta = -2^{12}3^6c^2(16+27c^2) \neq 0$. This certifies smoothness. At $X = -4$ the right side is $-432c^2$, so this exceptional denominator has no real zero on E_c.

The three points at infinity of the plane sharing cubic are smooth. They correspond to O and $(-4,\pm 12c\sqrt{(-3)})$. The functions u,v have simple poles there; each scalar slot r,b,D has pole order three at each and hence degree nine. Only O is real. A function's degree is the number of generic complex preimages, equivalently its total pole order.

PROVED / TC-G2: keeping both rows is an extra condition

The row difference $r-b = d$ determines X. Distinct elliptic points with these rows must therefore be R and -R, and negation sends (u,v) to (-v,-u). Equality of r at those points is precisely

$$(u+v)(u^2-uv+v^2-1) = 0.$$

The first factor gives $Y = 0$ and $R = -R$, not two columns. The second factor, together with the sharing equation, gives

$$3d^2+\ell^2 = 4,\ 2d^3-4d+c = 0,\ \ell \neq 0.$$

The original real equation forces $d > 0$, and $\ell \neq 0$ gives $d < 2/\sqrt{3}$. The derivative of $4d-2d^3$ changes sign only once on that interval, so there are at most two d values. Exclude $d^2 = \ell^2 = 1$ to avoid equal row and column scalar sets. Put $h = d/2$ and $s = |\ell|/2$. The frame center is $q = (z+w)/4$ and its marked points are

$$R_\pm = (44-96h^2,\ \pm 288s(1-2h^2)),\ 3h^2+s^2 = 1.$$

Meaning → limitation. Each positive width has two column choices encoded by ±R, not two different frames. Fixed z,w give at most two widths/four points. Only fixing c leaves a free scalar translation of q. No arbitrary network classification follows.

The completion cover and its exact scope

PROVED / TC-G4: branch count before genus

Keep one row r with output z. Factoring $P(r,r-u) = P(r,r-u')$ gives $(u-u')(u^2+uu'+u'^2-1) = 0$. The nontrivial branch is

$$u' = (-u+W_{cov})/2,\ W_{cov}^{\ 2} = 4-3u^2.$$

The normalization D_c is the smooth projective curve obtained from this double cover of E_c. The map u has degree three and three simple poles. At either $u = \pm 2/\sqrt{3}$, v satisfies $v^3+v = u^3+u-c$. A repeated root would have $v = \pm i/\sqrt{3}$ and a nonreal critical value v^3+v, while the right side is real. Thus there are three distinct v values at each sign, and u is a local coordinate at all six points.

The radicand has six simple zeros and three poles of order two. A square of a rational function has only even zero and pole orders. Hence the radicand is not a square, the cover is connected, and precisely the six zeros are branch points. Riemann-Hurwitz now gives

$$2g(D_c)-2 = 2(2g(E_c)-2)+6 = 6.$$

For any degree-e isogeny $E' \to E_c$, followed if desired by a translation, characteristic zero makes the map unbranched. The six branch points pull back to 6e distinct simple zeros. Their odd orders keep the pulled-back double cover connected, and its genus is 1+3e.

What the scalar formulas do and do not preserve

$$r,\ D = r-u,\ D' = r-u',\ r' = r-u-u',$$

$$q' = r-(u+u')/2,\ h' = (u+u')/2,\ s' = (u'-u)/2.$$

These slots satisfy $3h'^2+s'^2 = 1$, with outputs z and $4q'-z$. Each slot has degree eighteen on D_c: there are six points above the old poles, and the cubic term of r gives pole order three at each. The linear subtractions do not cancel it. The second row need not be b, and the second output need not be w.

Real completion requires $|u| \le 2/\sqrt{3}$. Four distinct scalars require u, u′, u+u′ and u′-u all nonzero; poles are removed. These conditions list all pairwise differences. A zero value of W_{cov} gives one branch choice and does not alone force scalar coincidence.

The obstruction concerns rational functions, not isolated points

A nonconstant rational map between smooth projective curves extends over its missing points. A genus-one source mapping to D_c would violate Riemann-Hurwitz: $0 = 6m + R$ with $m \ge 1$ and $R \ge 0$. This excludes choosing the extra branch rationally on an elliptic source, including after an isogeny. It neither counts nor excludes isolated real/algebraic lifts. Imposing $r' = b$ and the original w instead recovers A1's separate finite-locus theorem.

The exact attachment, field and marked shift

PROVED / TC-G5 and MS-C1: fixed conventions

$$a = \sqrt{2},\ \lambda = (3+a)/14,\ H = \sqrt{\lambda} > 0,\ W = \sqrt{(4a-3)} > 0.$$

Here W is the fixed radical of TC-G5/MS-C1, not the variable cover coordinate W_{cov} in A2. Set $h_1 = (a-1)H$, $h_2 = H$, $s_1 = HW$, $s_2 = h_1$, $c = 8H(4-a)/7$ and $q = c/4$. Then $c^2 = 64(19-3a)/343$ and E_c is defined over $F_0 = Q(a)$.

The positive-Y marked points and their group difference are exactly

$$R_1 = ((68+144a)/7,\ 288HW(2+3a)/7),$$

$$R_2 = ((164-48a)/7,\ 288H(5a-6)/7),$$

$$X(T) = [110+102a+(54+18a)W]/7,$$

$$Y(T) = H[1368+540a+(792+684a)W]/7.$$

The chord through R_2 and $-R_1$ has slope $m_T = 3H[a-2-(a+2)W]$. The formulas $X(T) = m_T^2 - X(R_1) - X(R_2)$ and $Y(T) = m_T(X(R_2)-X(T)) - Y(R_2)$ give these signs. The preserved referee appendix records the full expansion.

Why the needed automorphism exists

Put $F = F_0(H)$, $L = F(W)$. For F_0, the quadratic norm multiplies the conjugates $a \mapsto \pm a$, so $N(r+sa) = r^2-2s^2$ for rational r,s. The source's square-class calculation is

$$N(\lambda) = 1/28,\ N(4a-3) = -23,$$
$$N((4a-3)/\lambda) = -644.$$

The first value is not a rational square, so $[F:F_0] = 2$. Neither of the last two values can be the norm of a square. If $4a-3 = (x+yH)^2$ with $x,y \in F_0$, the H coefficient forces $2xy = 0$, contradicting one of those two tests. Thus $[L:Q] = 8$. The point coordinates recover a, then H, then W; L is their exact joint coordinate field. No minimal-field claim for T alone is required.

$$\sigma(a) = a,\ \sigma(H) = H,\ \sigma(W) = -W,$$
$$T + \sigma(T) = 2R_2.$$

Meaning → limitation. This valid automorphism fixes R_2 and negates R_1. If T were torsion, $2R_2$ would be torsion. It reduces the one marked-point question to R_2, without a rank or whole-group classification.

The one-prime valuation certificate

PROVED / MS-C1: all arithmetic hypotheses at p = 23

For a rational number, v_p is the exponent of p in its factorization, normalized by $v_p(p) = 1$. Extend it to Q_p, the completion for the size p^{-v_p}. The ring Z_p consists of elements of valuation at least zero; its units have valuation zero. Reduction takes an integral element to its residue modulo p.

$$F \to Q_{23},\ a \equiv 18,\ H \equiv 6 \pmod{23}.$$

The root 18 of a^2-2 has nonzero derivative 36 modulo 23. After lifting a, $\lambda = (3+a)/14$ has residue 13, and 6 is a simple root of $H^2-\lambda$ with derivative 12. Simple-root lifting means that a root known modulo 23^n has a unique next digit cancelling its error modulo 23^{n+1}. The compatible digits converge in Z_{23}. They define the displayed field embedding. It need not respect the ordering of the original real field, and W need not embed in Q_{23}.

Good reduction means that the integral model remains nonsingular modulo 23. For this short equation it is certified by its discriminant being a unit:

$$\Delta = -2^{12}3^6c^2(16+27c^2),$$
$$c^2 \equiv 16,\ 16+27c^2 \equiv 11 \pmod{23}.$$

A fixed doubling, not an order search

The tangent slope at R_2 is H(16a-4). Applying the group law once gives the source identity

$$S = 2R_2 = (48+24a,\ -16H(281+208a)/7).$$

$$(281+208a)(281-208a) = -7567 = -7\cdot 23\cdot 47.$$

At the chosen embedding, 281+208a has residue 0 and 281-208a has residue 10. Since the latter is a unit and the product has valuation one, $v_{23}(281+208a) = 1$ exactly. The factors 16,H,7 are units, so $v_{23}(Y(S)) = 1$. Further, $X(S) \equiv -3$ and $3X(S)^2-48 \equiv 2$.

$$\mu = (3X(S)^2-48)/(2Y(S)),\ v_{23}(\mu) = -1,$$

$$U = 4R_2,\ X(U) = \mu^2-2X(S),\ v_{23}(X(U)) = -2.$$

The two terms of X(U) have valuations -2 and 0, so cancellation is impossible. The denominator is nonzero, so U is a finite point. The following local lemma excludes torsion at every possible order.

Scope. Exactly one prime is used. The factor 47 is part of an integer norm identity, not a second local test. This freeze reproduces the saved equalities; no arithmetic program or division search is run.

Why the formal logarithm excludes all orders

PROVED / supplied EG-C2 lemma, applied in MS-C1

Meaning and statement. Near O, the elliptic group can be mapped injectively enough for torsion detection into an additive field of characteristic zero. If $p \geq 5$ and $y^2 = x^3+Ax+B$ has $A,B \in Z_p$ and unit discriminant, every finite Q_p-point with $v_p(x) < 0$ has infinite order.

Proof. Use local coordinates $t = -x/y$ and $w = -1/y$ at O. The equation is $w = t^3+Atw^2+Bw^3$. Recursive coefficient comparison gives $w(t) \in t^3Z_p[[t]]$, with $w = t^3+O(t^7)$. The smooth integral group law has an integral local power series $F(t_1,t_2) = t_1+t_2+$ terms of higher degree. Thus $t \in pZ_p$ describes a subgroup near O.

An invariant differential is a differential unchanged by translation in the group. Here it is $dx/(2y)$; in t it has integral expansion

$$\omega = [(tw'(t)-w(t))/(2w(t))]\, dt$$
$$= (1 + \sum_{j\geq 1} a_j t^j)\, dt,\ a_j \in Z_p.$$

For completeness, invariance is verified in the supplied proof by translating with a fixed point (r,s): $m = (y-s)/(x-r)$, $x' = m^2-x-r$, $y' = -y+m(x-x')$. The identity $3x^2+A-2my = (x-r)(2x+r-m^2)$, followed by differentiation, gives $dx'/(2y') = dx/(2y)$. Regularity extends this across exceptional chord denominators.

The **formal logarithm** is the termwise integral of this power series, normalized to vanish at O:

$$\log_E(t) = t + \sum_{j\geq 1} [a_j/(j+1)]\, t^{j+1}.$$

If $v_p(t) = r \geq 1$, each later term has valuation at least $(j+1)r-v_p(j+1) > r$, and these bounds tend to infinity. Indeed, for $k = v_p(j+1)$, $p^k \leq j+1$ and $p^k \geq 1+(p-1)k$ give $k \leq j/(p-1) < j$. The series converges and its leading term cannot cancel: $v_p(\log_E(t)) = r$.

Invariance of ω gives $\log_E(F(t_1,t_2)) = \log_E(t_1)+\log_E(t_2)$. A nonzero torsion parameter would therefore give $n \log_E(t) = 0$ with $\log_E(t) \neq 0$ in characteristic zero, a contradiction.

Finally, if $v_p(x) = -b < 0$, x^3 strictly dominates Ax and B. Hence $2v_p(y) = -3b$; write $b = 2r$, so $v_p(t) = r \geq 1$. This is exactly the subgroup just treated. Applying the lemma to $U = 4R_2$ at $p = 23$ proves R_2, and then T by $T+\sigma(T) = 2R_2$, have infinite order.

Limitation. This is a local torsion criterion on the specified good model over Q_p. It is not a torsion classification over arbitrary ramified extensions. Source: EG-C2, reproduced with checked hypotheses in MS-C1 arithmetic §5 and the saved referee audit.

Source and status ledger

The following identifiers locate complete supplied arguments or retained finite evidence. The chronology is TC checkpoint, then MS certificate, then this documentation freeze. All companions required for this report are present. Older migration gaps are not used as proof dependencies.

Claim / frozen status	Meaning and precise source scope
TC-G1 / PROVED	One common column; geometry.md TC-G1 in source [TC].
TC-G2 / PROVED	At fixed z,w, at most two admissible widths/four real marked points; [TC], TC-G2.
TC-G4 / PROVED	General one-row completion has genus 4; isogeny pullback genus 1+3e; [TC], TC-G4.
TC-G5 / PROVED	Exact finite attachment; N=6, Q=8, Z=2, B=28; [TC], TC-G5.
TC-X1 / EXPERIMENTAL	Retained exact, exhaustive finite audit: full image 30; 28 singleton fibers and two size-4 fibers; [TC], exact-audit.md §4.
MS-C1 / PROVED	The specified T has infinite order; [MS], arithmetic.md §§1-5. Recorded independent internal review: referee-audit.md.
TC-O1 / RESOLVED	Former OPEN status in [TC] is historical and superseded by MS-C1. Earlier files remain unchanged.
T01 / OPEN	Arbitrarily large same-set real squares with a fixed positive image exponent saving. No construction is supplied.

Source locators, versions and evidence

[TC] research/notes/2026-10-11-torsion-compatible-sharing/. README.md, version 1; geometry.md, version 2; exact-audit.md and verification.md, version 1; exact-results.json. Titles: *Torsion compatible sharing and the finite frame obstruction* and its companions. The saved exact_audit.py is preserved and not executed.

[MS] research/notes/2026-10-11-marked-shift-certificate/. README.md, arithmetic.md, referee-audit.md and verification.md, all version 1. Title: *Infinite order certificate for the marked bridge shift*. The prior internal arithmetic and referee audits are source evidence, not external review and not a new audit by this freeze.

[EG] research/notes/2026-09-30-elliptic-geometry/complex-extensions.md, EG-C2 local formal-logarithm lemma. Its full argument is reproduced in [MS], arithmetic.md §5. Only this earlier local criterion is needed here.

Preservation and unresolved boundary

The standalone source-status-ledger.md supplies source links and SHA-256 hashes. Earlier proofs are preserved. This freeze checks exposition, availability and rendering only; it does not rerun the finite audit. The original cubic target remains OPEN. No successor task is adopted.